Retail Sales Analytics Using Exploratory Data Analysis, Association Rule Mining, and Machine Learning

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Retail sales transaction data provides valuable insights into customer purchasing behaviour and supports data-driven business decision-making. This study analyses a retail sales dataset containing 185,950 transactions recorded throughout 2019 to investigate sales patterns, identify frequently co-purchased products, and evaluate machine learning models for sales prediction. The analysis combines exploratory data analysis, Pearson correlation analysis, association rule mining using the Apriori algorithm, and three supervised machine learning models, including Linear Regression, Random Forest, and Extreme Gradient Boosting (XGBoost). Exploratory analysis revealed clear temporal purchasing patterns, substantial differences between sales volume and revenue contribution across products, and a negative relationship between product price and purchase frequency. Association rule mining identified strong purchasing relationships between smartphones and their compatible charging accessories, highlighting potential cross-selling opportunities. Among the predictive models evaluated, XGBoost achieved the best overall performance by providing the lowest prediction error while maintaining high computational efficiency. Feature importance analysis further demonstrated that product-related variables were the dominant predictors of sales revenue. Overall, the findings illustrate how integrating descriptive analytics, market basket analysis, and machine learning can transform retail transaction data into actionable business insights for inventory management, product recommendation, marketing strategy, and sales forecasting.

Keywords—Retail sales analytics, exploratory data analysis, association rule mining, Apriori algorithm, machine learning, XGBoost, sales prediction

I. Introduction

Sales transaction data contains critical insights related to the product sales, consumer behavior and company's operation. Analyzing historical sales data allows organizations to learn about consumer behavior, discover demands of consumers, and make data-driven business decisions. Apart from the descriptive analysis, machine learning methods help businesses predict sales results in the future [1], and the market basket analysis allows companies to learn which products are often purchased together and apply cross-selling strategies.

This project studies sales data of an electronic retailer, including the products sold, the value of sales, purchase time, and location of clients. The analysis involves several approaches, including exploratory data analysis, statistical analysis, mining association rules with the help of the Apriori algorithm, and supervised machine learning. There are three algorithms used in this project to develop predictive models, including linear regression, random forest, and extreme gradient boosting (XGBoost).

Objectives of this study include the following:

- Conducting exploratory data analysis for understanding past sales trends;
- determining which products tend to be purchased together using association rules mining; and
- assessing machine learning models for predicting future sales and selecting key predictor variables.

Results obtained in this research shed light on consumers' buying patterns and how data analysis and machine learning methods can help inform decisions in the business world.

II. Dataset Understanding and Preprocessing

A. Data Description

The dataset consists of **185,950 records** of retail sales transactions for **2019**. Each record refers to one sale, and it comprises details on the product, the quantity sold, the unit price, time of sale, sales revenue, purchase location, and the order ID.

The original variables and data types are as presented in Table I.:

TABLE I. Sales Data Overview

Row No.	***Variable***	***Datatype***
0	Unnamed:0	Int64
1	Order ID	Int64
2	Product	Str
3	Quantity Ordered	Int64
4	Price Each	Float64
5	Order Date	Str
6	Purchase Address	Str
7	Month	Int64
8	Sales	Float64
9	City	Str
10	Hour	Int64

B. Data Cleaning

Preprocessing steps were implemented before the actual analysis was done to ensure good-quality data.

Checks for missing values and duplicates were conducted and there were **no missing values and duplicated records** discovered.

The data type of *Order ID* was changed into *string* type since it acts as identifier rather than integer for mathematical computation. *Order Date* was changed to *datetime* data type for calculation and better understanding of time range of the data.

After examination on the distribution and duplicates of the *Unnamed:0* column, it can be inferred that this column provided no informative message for analysis and thus, was removed during the preprocessing.

C. Feature Engineering and Data Preparation

To prepare the dataset for machine learning:

- Identifier variable *Order ID* was removed before feeding data into the models； duplicate features like *Order Date* and *Purchasing Address* were dropped as attributes like *Month, Hour, City* were extracted from them.
- *Price Each* was removed before model training since it exhibited perfect correlation with the target variable (*Sales*), as identified during correlation analysis.
- Categorical variables (*Product* and *City*) were encoded using dummy encoding technique [2].
- Numerical features such as *Quantity Ordered* and *Sales* were standardised before fitting Linear Regression to ensure comparable feature scales. Tree-based models (Random Forest and XGBoost) were trained using the original feature scales, as standardisation is not required for these algorithms [3].
- The data was randomly split into 70% training data and 30% testing data using a fixed random seed to ensure reproducibility.

III. Exploratory Data Analysis and Statistical Analysis

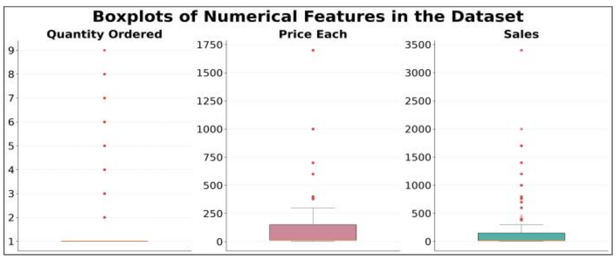
In order to have an overview and better understanding on key sales patterns and customer purchasing behaviors, the retail sales dataset was explored by exploratory data analysis (EDA) and statistical analysis. Visualisations and summary statistics are first used to examine data distributions, temporal sales trends, product performance, and the relationship between product price and purchase frequency. Pearson correlation analysis is then conducted to quantify relationships among numerical variables and support feature selection for the subsequent machine learning models. Together, these analyses provide both descriptive insights into historical sales performance and a statistical foundation for the market basket analysis and predictive modelling presented in the following sections.

A. Sales Distribution and Data Characteristics

An overview on distribution and statistics of the numeric variables of the dataset was investigated by boxplots and statistical description. *Month* and *Hour* were excluded as they are temporal cyclical.

Price Each and *Sales* are severely right skewed according to Fig. 1, while *Quantity Ordered* mainly concentrate at 1. **This means most customers purchase single product per order at lower prices.**

Fig. 1. Boxplots of Numerical Features in the Dataset



With details shown in Table II., 75% of the products were priced less than $150. Several extremely high-valued (e.g. $1700 per each) products contributed as outliers. These observations were considered as genuine purchases of expensive products rather than data quality issues and were therefore retained for subsequent analysis.

TABLE II. Statistics of Numerical Features in the Dataset

Statistics	Variables		
	Quantity Ordered	***Price Each***	***Sales***
Count	185950.00	185950.00	185950.00
Mean	1.124	184.40	185.49
Min	1.00	2.99	2.99
25%	1.00	11.95	11.95
Median	1.00	14.95	4.95
75%	1.00	150.00	150.00
Max	9.00	1700.00	3400.00
Std	0.44	332.73	332.92

B. Sales Patterns

- Monthly revenue: Fig. 2 indicates that sales revenue fluctuated across the year, with **December generating the highest sales (USD 4.61 million)**.

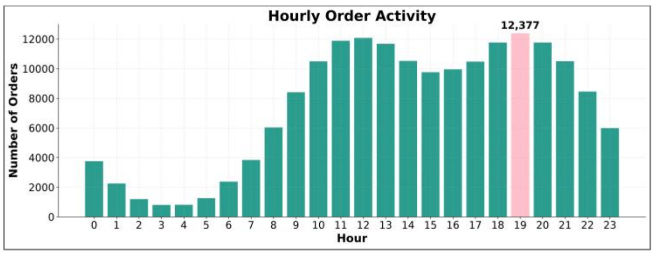
Fig. 2. Monthly Sales

- Sales by city: Overall, **San Francisco contributed the most sales revenue,** with around USD 8.26 million as presented by Fig. 3.

Fig. 3. Sales by City

- Order activity: Order activity gradually increased throughout the morning time. **The most purchase happened at 19:00, reaching 12,377 orders** (Fig. 4)**.**

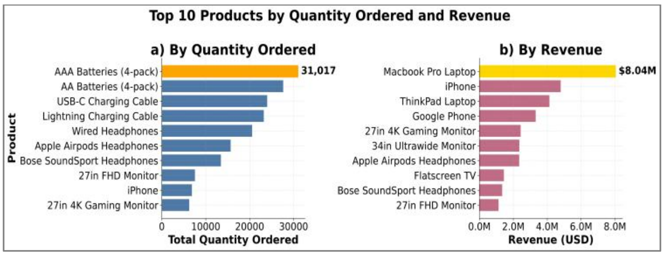
Fig. 4. Hourly Order Activity



C. Product Performance

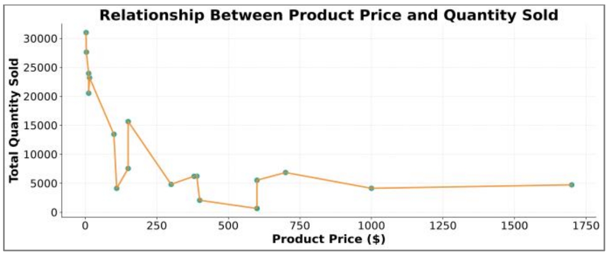
Fig. 5 compares the top 10 products ranked by total quantity ordered and total revenue. From Fig. 5(a), it is apparent that **AAA Batteries (4-pack) topped in terms of sales quantity at 31,017 units,** while AA Batteries (4-pack) and charging cables were the next. These frequently consumed products contributed to large sales volumes. On the other hand, **MacBook Pro Laptop generated the highest sales revenue at about USD 8.04 million** despite not being included in the list of the top-selling products by quantity, as illustrated in Fig. 5(b). The same case applies to the iPhone and ThinkPad Laptop which are the expensive devices. This observation suggests a potential inverse relationship between product price and purchase frequency, which is further investigated in the following analysis.

Fig. 5. Hourly Order ActivityTop10 Products by Quantity Ordered and Revenue



Examination on relationship between product price and quantity ordered through scatter plot (Fig. 6) revealed that **lower-priced products generally achieve higher sales volumes, whereas higher-priced products tend to sell in smaller quantities.** Though a few products deviate from this overall pattern, these exceptions likely reflect differences in product category and customer demand rather than contradicting the general trend.

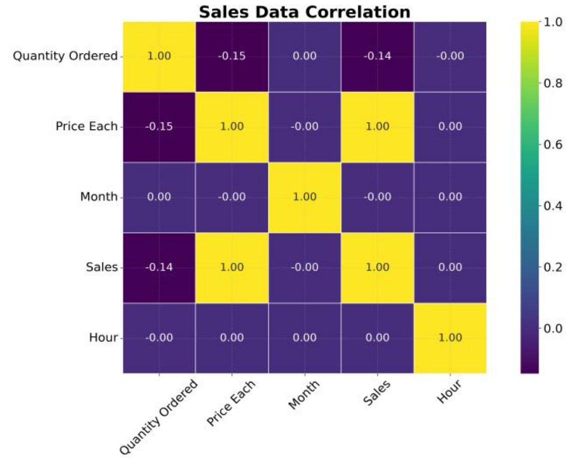
Fig. 6. Relationship Between Product Price and Quantity Sold



D. Statistical Relationships

To prepare features for the upcoming machine learning task, correlation among the numeric variables was investigated through Pearson correlation analysis. Findings are shown in Fig. 7 below.

Fig. 7. Sales Data Correlation Heat Map



Most of the features have weak linear correlations among themselves, which indicates that each feature is measuring unique things when it comes to the customer purchase behavior. The most correlated features in this case is ***Price Each* and *Sales* (Pearson correlation coefficient = 1.00),** and this is to be expected since the sales value for each transaction can be determined by equation (1). To avoid redundancy and data leakage in the upcoming machine learning models, *Price Each* feature was removed, while *Sales* was kept as the target variable.

$$Sales = Price\ Each \times Quantity\ Ordered \tag{1}$$

IV. Association Rule/Market Basket Analysis

Association rule mining was performed using the Apriori algorithm to investigate product bundling. This could provide practical insights for real business. Products frequently purchased together can be displayed together in store or recommended through e-commerce platforms. Cross-selling opportunities can be increased by selling these products in promotional package.

A. Transaction Characteristics and Frequent Itemsets

To first address which items were often bought together, transaction-level data were created by grouping products into baskets using *Order ID*. Before applying association rule mining, the distribution of basket sizes was examined to understand the characteristics of customer purchasing behaviour.

TABLE III. Summary of Basket Size

Number of Products in combination	*Number of Orders*	*Percentage*
1	171605	96.17
2	6479	3.63
3	337	0.19
4	15	0.01
5	1	0.00

As demonstrated in Table III., **single-product purchase dominated the orders with a percentage of 96.17% of the total transactions.** Only 3.83% orders included two or more products. It was very rare to have orders of three or more products, which amounted to less than 0.2% of all transactions.

Since association rule mining is based on the presence of items within the same transaction, the limited presence of transactions with more than one item makes it hard to find frequent itemsets. [4] In order to solve this problem, **a very low minimum support threshold shall be adopted later in the application of the Apriori algorithm.**

B. *Frequent Itemset Mining*

To identify product sets frequently purchased together, the Apriori algorithm was implemented on transaction dataset. The Apriori algorithm finds the frequent itemsets by looking for those product combinations which meet the minimum support value predefined. The support value is defined as the share of transactions that include a specific combination of products [4].

Various values for the support threshold were tested to reach a compromise between the number of the itemsets kept and their reliability.

TABLE IV. COMPARISON OF FREQUENT ITEMSET OUTPUTS BY DIFFERENT MINIMUM SUPPORT THRESHOLDS

Number of Total Transactions = 178,437			
Min_Support	***Min_Orders _Required***	***Frequent_Itemsets***	***Multi_Item _Itemsets***
0.0100	1785	17	0
0.0050	893	19	2
0.0010	179	27	8
0.0005	90	35	16
0.0002	36	55	36
0.0001	18	96	77

As indicated in Table IV., support thresholds above 0.005 provided insufficient number of multi-item itemsets, limiting the ability to discover meaningful purchasing relationships. In contrast, the support thresholds lower than 0.0005 produced more itemsets but also lowered the minimum occurrence requirement, increasing the possibility of random occurrence of that itemset. Hence, **0.0005 was selected as the more conservative threshold,** which requires each itemset to occur in at least 90 of the 178,437 orders while retaining 16 multi-item combinations. This help to eliminate chances of random purchasing and provide enough itemsets to study on.

C. *Association Rule Analysis*

Association rules were then generated from the selected frequent itemsets derived from the minimum support threshold of 0.0005. Three measures, **support, confidence and lift,** were applied to evaluate the rules. Support is a measure of the frequency of occurrence of a product combination in the dataset. Confidence on the other hand is the conditional probability of a consequent purchase given the anteceden. Lift is an indication of the occurrence of two products together more often than randomly expected [4]. A Lift value greater than one indicates a positive association between products.

TABLE V. SUMMARY OF ASSOCIATION RULE ANALYSIS

Antecedent	***Consequent***	***support***	***confidence***	***lift***
Google Phone	USB-C Charging Cable	0.0056	0.1806	1.4741
USB-C Charging Cable	Google Phone	0.0056	0.0456	1.4741
USB-C Charging Cable	Vareebadd Phone	0.0021	0.0168	1.4550
Vareebadd Phone	USB-C Charging Cable	0.0021	0.1782	1.4550
iPhone	Lightning Charging Cable	0.0057	0.1478	1.2208
Lightning Charging Cable	iPhone	0.0057	0.0468	1.2208

It can be summarised from Table V. that **charging accessories are the products most frequently purchased together with smartphones.** The most common combinations were USB-C Charging Cable and Google Phone (support = 0.56%, lift = 1.47), USB-C Charging Cable and Vareebadd Phone (support = 0.21%, lift = 1.45), and Lightning Charging Cable and iPhone (support = 0.57%, lift = 1.22). These combinations have lift values greater than 1, indicating that customers were more likely to purchase these products together than would be expected by chance.

Though the support values are relatively low (below 1%), this is expected because over 96% of orders in the dataset contained only one single product.

Overall, the results suggested that **customers commonly purchase compatible charging accessories together with smartphones,** highlighting a clear cross-selling opportunity for accessory recommendations.

V. MACHINE LEARNING MODELLING

Machine learning algorithms were employed to forecast sales revenues and extract the most influential in making the prediction. The dataset was first converted into a modelling dataset after selecting and processing the features. Three supervised machine learning models, Linear Regression, Random Forest, and XGBoost, were built and evaluated using standard regression metrics. Lastly, feature importance analysis was performed to interpret the contribution of individual predictors and provide insights into the factors influencing sales prediction.

A. *Feature Selection and data preparation*

As explained in Section II, the cleaned dataset was used for modelling after feature engineering and preprocessing. The final feature set consisted of three numerical variables (*Quantity Ordered, Month, and Hour)* and two categorical variables (*Product and City*), which were encoded using *Pandas* dummy encoding. *Sales* was used as the prediction target (see Table VI.).

TABLE VI. FEATURES USED FOR MACHINE LEARNING MODELS

Target Variable	*Predictors*	*Type*	*Standardisation*
Sales	Quantity Ordered	Numeric	Yes for Linear Regression model
	Month	Numeric	Yes for Linear Regression model
	Hour	Numeric	Yes for Linear Regression model
	Product	Categorical Dummy encoded)	/
	City	Categorical (Dummy encoded)	/

The dataset was randomly split into 70% training and 30% testing sets using a fixed random seed to ensure reproducibility.

For Linear Regression model, numerical variables were standardised prior to model fitting because the algorithm is sensitive to differences in feature scales [1]. In contrast, Random Forest and XGBoost were trained using the original feature values, as tree-based models are insensitive to feature scaling [1].

B. *Model Development and Performance Evaluation*

Three regression algorithms have been implemented to illustrate various modelling techniques used in prediction of sales revenue. Linear Regression was chosen as an easily understandable baseline model. The Random Forest algorithm has been used as an example of bagging technique for modelling nonlinear relationships [5], while XGBoost was selected as a gradient boosting algorithm due to its strong predictive performance and widespread use in structured tabular data [6]. Comparison of these models allows estimating both their prediction accuracy and computational efficiency.

All three models have been measured with *RMSE, R^2*, and runtime metrics.

TABLE VII. MODEL PERFORMANCE COMPARISON

Model	*RMSE*	*R^2*	*Runtime(seconds)*
Linear Regression	12.054	0.999	0.2037
Random Forest	1.795	1.000	41.9235
XGBoost	1.342	1.000	4.3725

As illustrated in Table VII, **all three regression models achieved excellent predictive performance,** with R^2 values close to 1.0. This means that the selected features explained almost all of the variation in sales revenue. Nevertheless, differences in prediction accuracy and computational efficiency were observed across the models. Linear Regression served as a strong baseline (R^2 = 0.999), but produced the highest prediction error (RMSE = 12.054). In contrast, the two tree-based ensemble models substantially reduced prediction errors, with Random Forest achieving an RMSE of 1.795 and XGBoost further reducing the RMSE to 1.342, the lowest among all models. In terms of computational efficiency, Random Forest spent the longest training time (41.92 seconds), whereas XGBoost completed training in only 4.37 seconds while maintaining the best predictive accuracy. These results suggest that **XGBoost provided the best balance between prediction accuracy and computational efficiency.**

Fig. 8. Model Performance Comparison - Predicted v.s. Actual

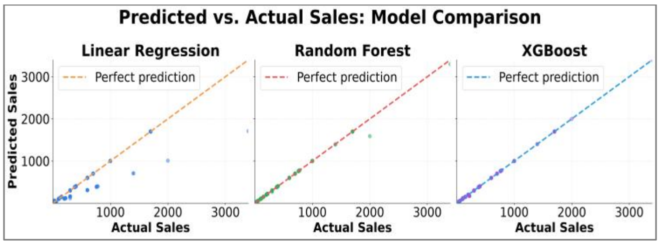


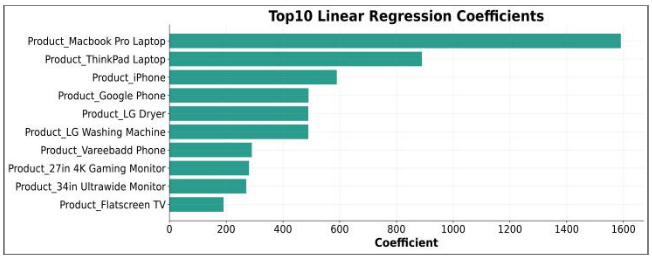
Fig. 8 further supports these findings by comparing the predicted and actual sales values for the three models. The predictions generated by Linear Regression generally followed the 45-degree reference line but exhibited noticeably larger deviations for several high-sales observations, indicating reduced accuracy at the upper end of the sales range. Random Forest produced predictions that clustered much more closely around the reference line, reflecting a substantial improvement in prediction accuracy. Among the three models, XGBoost demonstrated the closest agreement between predicted and actual values, with almost all observations lying on or very near the ideal prediction line. This visual evidence is consistent with the evaluation metrics in Table VII, confirming that **XGBoost generalised most effectively to the testing data and therefore provided the most reliable sales predictions.**

C. *Feature Importance Analysis*

Since XGBoost achieved the best predictive performance among the three models, its feature importance was used as the primary basis for interpreting which variables contributed most to sales prediction. Feature importance from Linear Regression and Random Forest was also presented for comparison to evaluate the consistency of the findings across different modelling approaches.

Although there were differences in the algorithms, all three models consistently identified **product-related variables as the dominant predictors of sales revenue** as demonstrated by Fig. 9, Fig. 10 and Fig. 11.

Fig. 9. Top10 Linear Regression Coefficients



According to Fig. 11, *MacBook Pro Laptop* was the most important variable, then came *ThinkPad Laptop*, *iPhone,* and *Google Phone.* The abrupt drop in importance after the first four products implied that sales revenue prediction is mainly driven by the product category rather than the time (*Month, Hour*) and location (*City*) of sales.

Fig. 10. Top10 Important Features - Random Forest

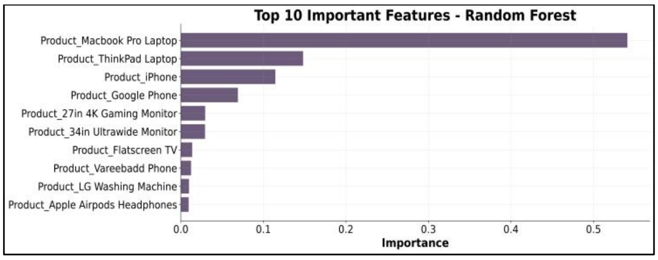
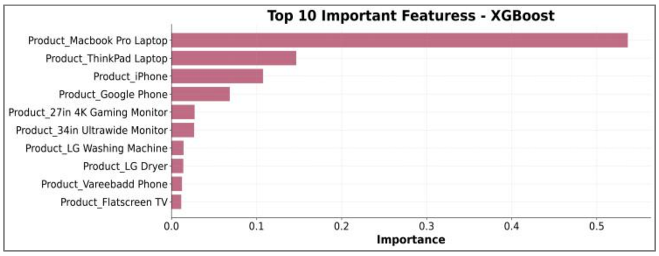


Fig. 11. Top10 Important Features - Random Forest



Similar patterns were observed in both Linear Regression coefficients (Fig. 9) and Random Forest feature importance (Fig. 10), where premium laptop and smartphone products consistently appeared among the most influential variables. This consistency increases confidence that the identified features genuinely contribute to sales prediction rather than being artefacts of a particular modelling algorithm.

By referring to the EDA, the dominance of product-related variables is actually expected as each product has a fixed selling price. Consequently, product identity implicitly captures much of the pricing information, making it the strongest predictor of sales revenue.

Overall, these findings **suggest that businesses should prioritise inventory planning and promotional activities for high-value products.** Combined with the association rule analysis, the results also highlighted opportunities for cross-selling complementary accessories with premium electronic devices.

VI. Discussion

This study employed exploratory data analysis, association rule mining, and machine learning methods to analyze sales trends and make predictions for sales revenue. The EDA revealed clear temporal purchasing patterns, product performance differences, and pricing characteristics across the dataset. Association rule analysis further identified several meaningful co-purchasing relationships, while the machine learning models demonstrated that sales revenue could be predicted with very high accuracy. Together, these findings provided a comprehensive understanding of customer purchasing behaviour and the key drivers of sales performance.

One important price-wise insight is that **high-value electronic products, particularly laptops and smartphones, contribute disproportionately to total revenue** despite selling in lower volumes than inexpensive accessories. This advises that revenue growth depends more on maintaining sales of premium products than simply increasing the volume of low-priced items.

On the other hand, **inexpensive accessories such as batteries and charging cables dominate total sales volume.** These products generated frequent transactions and may serve as entry-point purchases at the entrance or check-out display that encourages customer engagement.

In terms of temporal patterns, customer purchasing activity follows a clear daily trend, with **order volumes peaking during the evening.** This information could assist business in scheduling promotional campaigns, customer support resources, and inventory replenishment more effectively.

From product perspective, the association rule analysis further revealed that **smartphones are frequently purchased together with compatible charging accessories.** This confirms the existence of complementary purchasing behaviour and provides opportunities for product bundling and cross-selling strategies.

Based on the findings above, **it is recommended for business to prioritise inventory management for premium products,** as stock shortages of these items would have a critical impact on total revenue; **bundle promotions pairing smartphones with compatible charging accessories** could increase average transaction value while improving customer convenience; **marketing campaigns could be scheduled during peak purchasing periods** to maximise customer reach and conversion rates; for companies with sufficient computational investment, it is also suggested that **machine learning could be an effective decision-support tool to help predict revenue and plan on inventory management.**

To summarise, the findings in this report demonstrated how combining descriptive analytics, association rule mining, and predictive modelling provides complementary perspectives that support both operational decision-making and long-term business planning.

VII. Conclusion

This study analysed a retail sales dataset using exploratory data analysis, statistical analysis, association rule mining, and supervised machine learning techniques to investigate customer purchasing behaviour and predict sales revenue. The analysis identified clear temporal sales patterns, differences in product performance, and meaningful co-purchasing relationships. **Among the three predictive models evaluated, XGBoost achieved the best overall performance, demonstrating excellent predictive accuracy while maintaining reasonable computational efficiency.**

The findings suggested that combining descriptive analytics with predictive modelling provides valuable support for business decision-making. Exploratory analysis and market basket analysis revealed insights into customer behaviour that can inform inventory management, marketing strategies, and cross-selling opportunities, while the machine learning models demonstrated the potential to accurately forecast sales using transaction-level information.

Several limitations should also be acknowledged. The dataset contains only one year of historical transactions and does not include external factors such as promotions, seasonality across multiple years, customer demographics, or competitor activities that may influence purchasing behaviour. Furthermore, because each product is associated with a fixed selling price in this dataset, product identity becomes the dominant predictor of sales revenue, limiting the generalisability of the predictive models. Future work could incorporate additional years of data, customer-level information, promotional events, and dynamic pricing variables to develop more robust forecasting models and further improve business insights.

Overall, this project demonstrated how integrating data exploration, association rule mining, and machine learning can transform retail transaction data into actionable business knowledge, supporting more informed operational and strategic decision-making.

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